An Experimental Evaluation of AWS Cloud Security Misconfigurations Using Native Security Controls, Python, and SIEM-Based Log Analysis

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Abstract

Cloud security weaknesses can arise when identities, permissions, and monitoring controls are not configured or reviewed appropriately. This study evaluates a layered approach that combines AWS-native security controls, Python/Boto3 automation, AWS CloudTrail, Elasticsearch, and Kibana. A controlled AWS environment was used for configuration assessment, while a historical CloudTrail dataset was analyzed for behavioral evidence. The Kibana investigation contained 24,984 CloudTrail documents, including 1,700 IAM-related documents. Of the IAM records, 226 contained an error and 164 contained an AccessDenied result. Focused analysis identified denied attempts to create access keys, denied IAM-user creation, a denied role-policy attachment, and successful policy administration. The investigation also identified duplicate event identifiers, requiring raw document counts to be distinguished from unique events. The findings support the central proposition that configuration assessment and behavioral monitoring answer different security questions. Configuration assessment identifies security posture, while CloudTrail analysis provides evidence about actions attempted, identity, time, source, user agent, and outcome. The study therefore supports a layered security model rather than reliance on a single detection mechanism.

Keywords: AWS cloud security; IAM; CloudTrail; Python/Boto3; SIEM

1. Introduction

Cloud computing gives organizations access to computing, storage, networking, and application services without requiring them to operate all underlying infrastructure. This flexibility also changes the security problem. Cloud resources are controlled largely through identities, policies, APIs, and service configurations, so a security weakness can result from an authorization decision or configuration that is technically functioning but broader than the organization requires. NIST describes cloud security as involving security and privacy risks associated with externally provided computing resources, while the AWS shared-responsibility model places significant configuration and access-control responsibilities on the customer (Jansen & Grance, 2011; Amazon Web Services [AWS], 2026d).

Identity and Access Management (IAM) is central to this study because IAM policies determine which identities can perform AWS API operations. Excessive permissions can increase the impact of compromised credentials or misuse by an authorized identity. AWS recommends least privilege, multi-factor authentication, credential management, and regular permission review (AWS, 2026a, 2026b). NIST Cybersecurity Framework 2.0 similarly identifies the management and review of access permissions and authorization according to least privilege as a protection outcome (National Institute of Standards and Technology [NIST], 2024).

Configuration assessment alone, however, does not show how an identity behaves after the assessment is completed. CloudTrail provides a different type of evidence by recording AWS API activity, including the identity, event name, time, source information, user agent, request information, and outcome. When these records are centralized in a search and visualization platform, analysts can move from broad event volume to focused investigation.

This research evaluates a layered approach consisting of AWS-native controls, Python/Boto3 configuration assessment, and CloudTrail analysis through Elasticsearch and Kibana. The objective is not to prove that one tool detects every cloud threat. Instead, the objective is to determine whether the layers provide complementary visibility and whether the resulting evidence supports defensible security investigation.

2. Research Problem and Questions

The research problem is the gap between static knowledge of cloud security posture and behavioral evidence about what identities actually attempt to do. Manual configuration reviews can identify selected weaknesses, but they are snapshots. Conversely, a CloudTrail dataset may contain thousands of routine events, making broad log review inefficient. A practical security process therefore needs a repeatable way to assess configuration and a method for investigating relevant activity.

The primary research question is: How effective is a combined approach using AWS-native security controls, Python/Boto3 automation, and SIEM-based CloudTrail analysis for identifying AWS cloud security weaknesses and security-relevant IAM activity?

Three supporting questions guide the study:

- Can AWS-native security controls provide a practical baseline for selected IAM security conditions?
- Can Python/Boto3 make selected configuration checks repeatable and suitable for recurring assessment?
- Can CloudTrail and Elasticsearch/Kibana identify security-relevant IAM activity that a configuration assessment alone would not reveal?

For this study, effectiveness is defined narrowly as the ability to identify relevant conditions, produce repeatable evidence, support investigation and correlation, and expose limitations that require human analysis. Effectiveness is not defined as perfect detection, proof of malicious intent, or a measured false-positive rate. Those broader measures require experiments that were outside the scope of the available laboratory data.

3. Background and Related Work

Cloud Security and Shared Responsibility

Cloud security is influenced by both provider controls and customer decisions. NIST emphasizes that public-cloud use introduces security and privacy considerations that organizations must manage, while AWS describes a shared-responsibility model in which AWS secures the underlying infrastructure and customers remain responsible for many controls within their environments (Jansen & Grance, 2011; AWS, 2026d). This distinction supports the study's focus on IAM, configuration, logging, and monitoring.

Cloud Misconfiguration and IAM

Cloud misconfiguration can include excessive permissions, unnecessary credentials, weak authentication settings, or security controls that are not aligned with organizational requirements. Alouffi et al. (2021) identify multiple cloud-security threats and emphasize that mitigation requires complementary controls rather than reliance on a single mechanism. IAM is particularly important because an identity's permissions can affect multiple resources and services.

Least privilege limits permissions to those required for an authorized task. AWS recommends least privilege and periodic permission review, while NIST CSF 2.0 identifies

access permissions, entitlements, and authorizations as outcomes that should be managed, enforced, and reviewed using least-privilege principles (AWS, 2026b; NIST, 2024). MFA strengthens authentication but does not by itself establish that an authenticated identity has appropriate authorization. This distinction is important because the study examines both authorization configuration and observed API activity.

Automation and Behavioral Monitoring

Python and Boto3 provide programmatic access to AWS services and can support repeatable security checks. Automation is valuable because the same documented rule can be executed consistently. Its limitation is context: a script can identify that a permission or credential exists, but it may not know whether that object is required for a legitimate business function. Automated assessment therefore supports, rather than replaces, analyst judgment (AWS, 2026e).

CloudTrail adds behavioral evidence. Instead of asking only what permissions exist, an analyst can ask which identity attempted an operation, when it occurred, where the request originated, what client or user agent was used, and whether AWS allowed the operation. Elasticsearch and Kibana provide a practical environment for filtering and correlating these records.

Research Gap

The gap addressed by this study is practical rather than theoretical. Configuration assessment and log analysis are often treated as separate activities. This study places them in one evaluation framework and examines how their evidence differs. The contribution is therefore a defensible demonstration that posture evidence and behavioral evidence should be correlated rather than substituted for one another.

4. Threat and Security Gap Analysis

The threat model focuses on IAM-related weaknesses and activity that could increase the opportunity for unauthorized access or privilege escalation. The study does not assume that every unusual event is an attack. Instead, it identifies events that have security significance and evaluates their context.

Threat or Gap	Evidence of Concern	Relevant Control
Excessive IAM permissions	Identity can perform unnecessary actions	Least privilege; permission review
Credential misuse	Access-key creation or other credential activity	MFA; credential controls; CloudTrail
Unauthorized identity creation	CreateUser activity	IAM authorization; monitoring
Privilege modification	Policy attachment or policy-version changes	Least privilege; change control; logging
Detection blind spots	High event volume obscures relevant events	CloudTrail + SIEM filtering

The most important gap is the difference between configuration and behavior. A configuration scanner can indicate that a policy is attached to a user. It cannot, by itself, establish when that identity attempted a sensitive operation or whether AWS denied the request. Conversely, CloudTrail can show an attempted operation but cannot, by itself, determine whether the underlying permission is justified by the organization's business requirements.

5. Methodology and Approach

Experimental Environment

Component	Purpose
AWS IAM	Identity and permission management

AWS EC2	Controlled cloud test environment
AWS CloudTrail	Collection of API activity
Python/Boto3	Repeatable configuration assessment
Elasticsearch 9.5.4	Log storage and search
Kibana 9.5.4	Visualization and investigation
Windows 11	Local analysis workstation

The study used two related but distinct components. The first was a controlled AWS environment used to examine configuration and IAM concepts. The second was the supplied aws-Lab04 historical CloudTrail dataset used for behavioral analysis. The components were kept analytically separate because a configuration assessment measures security posture at a point in time, while the historical dataset measures recorded activity over a period.

Configuration Assessment

The Python/Boto3 layer was designed around repeatable checks of IAM users, roles, policies, credentials, and selected security settings. The intended workflow is to retrieve configuration attributes, compare them with defined security expectations, and report the resource, observed condition, expected condition, and recommended action. The study does not claim that the available script constitutes a complete AWS security scanner or that it produced a validated detection-rate benchmark.

CloudTrail and Kibana Procedure

The behavioral analysis began with the full dataset and progressively narrowed the search. The first step established the total document count. The next searches filtered for IAM events, IAM errors, AccessDenied results, and selected security-sensitive event names. Additional searches examined CreateAccessKey, CreateUser, AttachRolePolicy, AttachUserPolicy, PutUserPolicy, CreatePolicy, and GetPolicyVersion activity. The

MySecurityAudit policy sequence was investigated using event time, identity, event name, policy ARN, source IP, user agent, and outcome.

A key data-quality rule was applied after duplicate event identifiers were observed. Raw indexed documents were not automatically treated as unique events. The eventID field was used to distinguish repeated documents from individual CloudTrail events. This prevents inflated document counts from being interpreted as equivalent to incidents.

Evidence Interpretation and Evaluation

The study used conservative interpretation rules. A successful administrative action was not automatically classified as malicious. A denied action was not automatically classified as an attack. A Linux or Kali user agent was treated as contextual evidence, not proof of malicious intent. The four evaluation criteria were: identification of relevant security conditions, repeatability of evidence, support for investigation and correlation, and identification of limitations.

6. Findings and Discussion

6.1 Dataset and IAM Activity

The Kibana analysis identified 24,984 CloudTrail documents covering approximately February 16 through April 2, 2019. IAM filtering identified 1,700 documents, or approximately 6.80% of the dataset. Of the IAM documents, 226 contained an error and 164 contained an AccessDenied result. Thus, AccessDenied represented approximately 9.65% of IAM documents and 0.66% of all documents.

These numbers should not be interpreted as attack counts. CloudTrail contains normal administration, automation, service activity, and failed requests. The result demonstrates the

value of filtering: a large dataset can be reduced to a security-relevant subset before event-level investigation begins.

6.2 Security-Sensitive IAM Operations

Operation	Unique Events	Observed Outcome
CreateAccessKey	2	Both denied
CreateUser	2	Both denied
AttachRolePolicy	1	Denied
AttachUserPolicy	1	Successful
PutUserPolicy	1	Failed; policy-size limit
CreatePolicy	1	Successful

The focused search returned 14 raw documents, but duplicate event identifiers were present. The analysis therefore identified the unique events shown above. This distinction is a meaningful methodological finding because a security analyst who counts indexed documents without checking event identity could overstate activity.

6.3 Credential and User-Creation Attempts

Two unique CreateAccessKey events occurred on February 23, 2019. The Level6 identity attempted to create access keys associated with the backup and piper users, and both operations were denied. Two unique CreateUser events were also identified. One involved an attempted creation of bugteeth by Level6, and another involved an attempted creation of hugo by backup. Both were denied.

These events demonstrate security-relevant behavior because credential creation and user creation can materially change an AWS identity environment. However, the evidence does not establish why the requests were made. The defensible conclusion is that unauthorized-by-policy attempts were recorded and prevented, not that a compromise was proven.

6.4 MySecurityAudit Policy Sequence

The March 3 sequence illustrates why temporal correlation is important. At 09:39:31 UTC, a PutUserPolicy request targeting Level6 failed because the policy exceeded the 2,048-byte limit. At 09:42:45 UTC, a root identity successfully created the managed policy MySecurityAudit. At 09:43:14 UTC, the policy was successfully attached to Level6. The policy description indicated an audit-oriented purpose, and the root session was MFA-authenticated.

The sequence should therefore be interpreted as administrative or security-audit configuration activity rather than assumed malicious behavior. The important analytical result is not that the activity was suspicious; it is that Kibana allowed the analyst to correlate a failed attempt, a successful policy creation, and a subsequent attachment within minutes. A configuration snapshot alone would not provide that temporal sequence.

6.5 Denied Role-Policy Attachment and Subsequent Policy Access

On March 26, Level6 attempted an AttachRolePolicy operation involving SecurityAudit, and AWS returned AccessDenied. The record included source and client information associated with an AWS CLI session on Linux/Kali. Later that day, Level6 retrieved version v1 of MySecurityAudit using GetPolicyVersion. Again, the Kali user agent is not proof of malicious activity because the platform can be used legitimately for security testing and administration.

The sequence demonstrates the value of combining outcome and context. The denied policy attachment shows that authorization controls prevented the requested change. The later policy retrieval shows activity involving an audit policy. Neither event independently establishes compromise.

6.6 Evaluation of the Research Question

The findings support the primary research question within the study's defined scope. First, configuration-oriented controls provide a way to assess the security posture of identities and resources. Second, Python/Boto3 provides a repeatable mechanism for selected configuration checks, although the study did not measure comprehensive detection accuracy. Third, CloudTrail and Kibana reveal behavioral information that a static configuration assessment does not provide, including attempted operations, outcomes, timing, source information, and user-agent context.

The strongest evidence for the combined approach is the difference between posture and behavior. The Level6 events demonstrate that an analyst can observe denied credential and user-creation attempts even though the requests did not result in successful changes. The MySecurityAudit sequence demonstrates that successful administrative changes can also be reconstructed. The two forms of evidence answer different questions and therefore complement each other.

6.7 Comparative Evaluation of Splunk and Kibana

The CloudTrail dataset was also analyzed using Splunk Enterprise as a second SIEM platform. Splunk ingested the Lab-CloudTrail JSON data into the cloudtrial_lab_1 index and extracted fields including eventName, eventSource, eventTime, sourceIPAddress, userIdentity, and requestParameters. The initial Splunk search returned 25,000 events.

The comparative filtering produced results that closely matched the existing Kibana analysis. Splunk identified 1,700 IAM events, 226 IAM events containing an errorCode, and 164 IAM events with errorCode=AccessDenied. These values match the corresponding Kibana results of 1,700 IAM documents, 226 IAM error documents, and 164 AccessDenied documents.

The matching results provide additional confidence that the principal IAM and authorization findings were reproducible across two SIEM platforms.

The two platforms also provided complementary investigative views. Kibana was used to reduce the historical CloudTrail dataset to IAM activity, identify authorization failures, and examine security-sensitive IAM operations. Splunk provided an additional event-frequency and identity-oriented view. Its analysis identified 410 distinct event names, with DescribeSnapshots (6,161), AssumeRole (3,630), and DescribeReservedInstancesOfferings (3,563) among the most frequent operations.

Splunk's identity analysis identified 22,914 IAMUser events, 1,358 AWSService events, 474 AssumedRole events, and 254 Root events. Root activity included operations such as ListMFADevices, ListAccessKeys, ListUsers, ListPolicies, GetPolicyVersion, and DetachUserPolicy. These results demonstrate the value of identity-type filtering, but Root activity was not interpreted as proof of compromise because the available evidence does not establish malicious intent.

The small difference between the overall Kibana and Splunk event counts should not be interpreted as a contradiction. Kibana reported 24,984 documents while Splunk reported 25,000 indexed events. The more important comparison is that the same IAM, error, and AccessDenied filters produced identical counts. This demonstrates that the analytical conclusions were not dependent on a single SIEM interface.

Table 5. Comparison of CloudTrail analysis results in Kibana and Splunk.

Analysis	Kibana	Splunk	Result
Total CloudTrail records	24,984	25,000	Small count difference
IAM records	1,700	1,700	Exact match

IAM records with errors	226	226	Exact match
AccessDenied records	164	164	Exact match

6.8 Limitations and Threats to Validity

Several limitations restrict generalization. The CloudTrail dataset is historical and covers a limited period. The analysis focuses primarily on IAM rather than the full AWS service population. The controlled laboratory environment is not equivalent to a production enterprise. The available evidence does not establish the business purpose of every identity or operation. The Python/Boto3 component demonstrates an assessment approach but does not provide a statistically validated detection benchmark. Finally, no ground-truth incident labels were available, so precision, recall, false-positive rate, and false-negative rate were not calculated.

These limitations do not invalidate the findings; they define what can responsibly be concluded. The study demonstrates analytical capability and complementary evidence, not universal detection effectiveness.

7. Proposed Mitigation, Security Approach, and Incident Response Strategy

7.1 Layered Security Architecture

Layer	**Purpose**
IAM hardening	Reduce unnecessary privilege and credential exposure
AWS-native controls	Establish authentication, authorization, and security baselines
Python/Boto3	Perform repeatable configuration assessments
CloudTrail	Record AWS API activity for investigation
Elasticsearch/Kibana	Search, correlate, visualize, and investigate events

Incident response	Validate, contain, investigate, recover, and improve

This layered model aligns with NIST CSF 2.0's emphasis on managing and reviewing authorization and access permissions according to least privilege. It also reflects the central finding of this study: no single layer answers every security question (NIST, 2024).

7.2 IAM and Monitoring Recommendations

- Apply least privilege to users and roles and review permissions periodically.
- Require MFA for privileged identities and use stronger controls for administrative access.
- Minimize long-lived access keys and monitor credential-related API operations.
- Monitor CreateAccessKey, CreateUser, AttachRolePolicy, AttachUserPolicy, PutUserPolicy, CreatePolicyVersion, SetDefaultPolicyVersion, and DeletePolicyVersion.
- Correlate sensitive operations with identity, source IP, user agent, time, and outcome.
- Retain CloudTrail evidence long enough to support investigation and change reconstruction.
- Schedule Python/Boto3 assessments and retain results so changes can be compared over time.

7.3 Incident Response Strategy

When a sensitive IAM event occurs, the first step should be identification: determine the identity, operation, target, timestamp, source, user agent, and result. The second step is validation: determine whether the activity corresponds to an approved administrator, automation process, testing activity, or change request. If unauthorized activity is confirmed, containment may include credential rotation, disabling affected credentials, removing unauthorized

permissions, or restricting the identity. Investigation should examine related events before and after the triggering event. Recovery should restore the approved configuration and verify that logging and IAM controls remain effective. Finally, lessons learned should be used to improve detection rules and configuration standards.

The response should remain evidence-driven. A denied event may warrant investigation without requiring emergency containment. A successful privilege-changing operation from an unexpected identity or source may warrant higher priority. This approach reduces both missed incidents and unnecessary escalation.

8. Conclusion and Future Directions

This study evaluated a layered approach to AWS cloud security using AWS-native controls, Python/Boto3, CloudTrail, Elasticsearch, and Kibana. The results show that configuration assessment and behavioral monitoring provide different forms of security visibility. Configuration assessment describes the security posture of identities and resources, while CloudTrail and Kibana provide evidence about actions attempted and their outcomes.

The 24,984-document dataset was reduced to 1,700 IAM events, including 164 AccessDenied records. Focused analysis identified denied credential creation, denied user creation, denied role-policy attachment, successful policy administration, and subsequent policy access. The investigation also identified duplicate event identifiers, demonstrating why raw document counts should not automatically be treated as unique events.

The study does not establish confirmed compromise or malicious intent. Instead, it demonstrates that a layered approach can identify security-relevant activity and provide the context necessary for human investigation. This is the most defensible conclusion supported by the available evidence.

Future work should intentionally introduce known IAM misconfigurations into a controlled environment and establish ground truth for each condition. That experiment could measure detection rate, false positives, false negatives, analyst investigation time, and the relative contribution of AWS-native controls, Python/Boto3, and SIEM analysis. A future comparison of Elasticsearch/Kibana and Splunk using the same dataset would also provide a stronger usability and detection evaluation.

AI Use Acknowledgment

Generative artificial intelligence tools were used during preparation of this research paper to assist with organization, brainstorming, grammar and clarity improvements, and interpretation of technical concepts. The author conducted and reviewed the AWS laboratory activities, Elasticsearch configuration, and Kibana/CloudTrail analysis. The author reviewed the generated material and is responsible for the accuracy of the methodology, evidence, findings, conclusions, and final submitted work.

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Appendix A. Evidence Summary

Evidence	Result	Interpretation
Total CloudTrail documents	24,984	Historical dataset
IAM documents	1,700	6.80% of dataset
IAM documents with error	226	13.29% of IAM records
IAM AccessDenied	164	9.65% of IAM records
AccessDeniedException	0	No matching records
Selected security-sensitive raw documents	14	Duplicates observed
CreateAccessKey	2 unique	Both denied
CreateUser	2 unique	Both denied
AttachRolePolicy	1 unique	Denied
AttachUserPolicy	1 unique	Successful
PutUserPolicy	1 unique	Policy-size failure
CreatePolicy	1 unique	Successful

The evidence was collected from the aws-Lab04 dataset during the October 2026 laboratory investigation. Raw document counts and unique-event observations are intentionally distinguished because duplicate eventID values were observed.

Appendix B. Kibana and Splunk Queries Used

Splunk Enterprise queries used for the comparative analysis:

Total indexed events: index=cloudtrial_lab_1 | stats count

IAM events: index=cloudtrial_lab_1 eventSource="iam.amazonaws.com" | stats count

IAM errors: index=cloudtrial_lab_1 eventSource="iam.amazonaws.com" errorCode=* | stats count

AccessDenied: index=cloudtrial_lab_1 eventSource="iam.amazonaws.com" errorCode="AccessDenied" | stats count

Event frequency: index=cloudtrial_lab_1 | stats count by eventName | sort - count

Identity types: index=cloudtrial_lab_1 | stats count by userIdentity.type | sort - count

Purpose	**Kibana query**
IAM activity	eventSource:iam.amazonaws.com
IAM errors	eventSource:iam.amazonaws.com AND errorCode:*
Access denied	eventSource:iam.amazonaws.com AND errorCode:AccessDenied
Selected sensitive IAM operations	eventSource:iam.amazonaws.com AND (eventName:CreateAccessKey OR eventName:CreateUser OR eventName:AttachRolePolicy OR eventName:AttachUserPolicy OR eventName:PutUserPolicy OR eventName:CreatePolicyVersion OR eventName:SetDefaultPolicyVersion OR eventName:DeletePolicyVersion)
CreateAccessKey	eventSource:iam.amazonaws.com AND eventName:CreateAccessKey
CreateUser	eventSource:iam.amazonaws.com AND eventName:CreateUser
AttachRolePolicy	eventSource:iam.amazonaws.com AND eventName:AttachRolePolicy
AttachUserPolicy	eventSource:iam.amazonaws.com AND eventName:AttachUserPolicy
PutUserPolicy	eventSource:iam.amazonaws.com AND eventName:PutUserPolicy
CreatePolicy	eventSource:iam.amazonaws.com AND eventName:CreatePolicy
MySecurityAudit access	eventSource:iam.amazonaws.com AND eventName:GetPolicyVersion AND requestParameters.policyArn:"arn:aws:iam::811596193553:policy/MySecurityAudit"

Appendix C. Recommended Evidence Screenshots

- Overall 24,984-document Discover view.
- 1,700 IAM-event view.
- 226 IAM-error view.

• 164 AccessDenied view.

• Selected security-sensitive IAM operations.

• CreateAccessKey results showing the two denied events.

• CreateUser results showing the two denied events.

• AttachRolePolicy result showing AccessDenied.

• MySecurityAudit policy sequence.

• GetPolicyVersion result for MySecurityAudit.

Appendix D. Experimental Environment

Figure C-5. Splunk total CloudTrail events: 25,000.

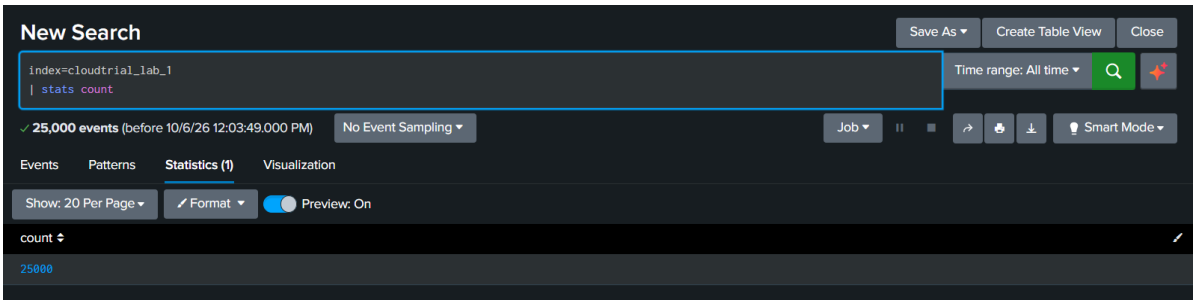


Figure C-6. Splunk IAM event count: 1,700.

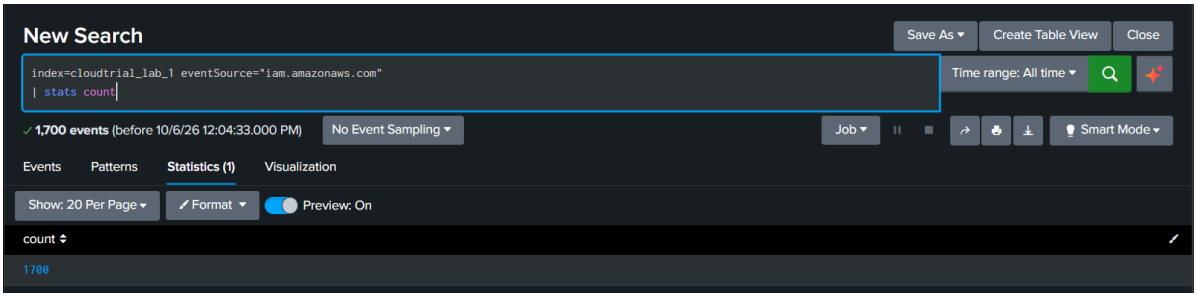


Figure C-7. Splunk IAM error event count: 226.

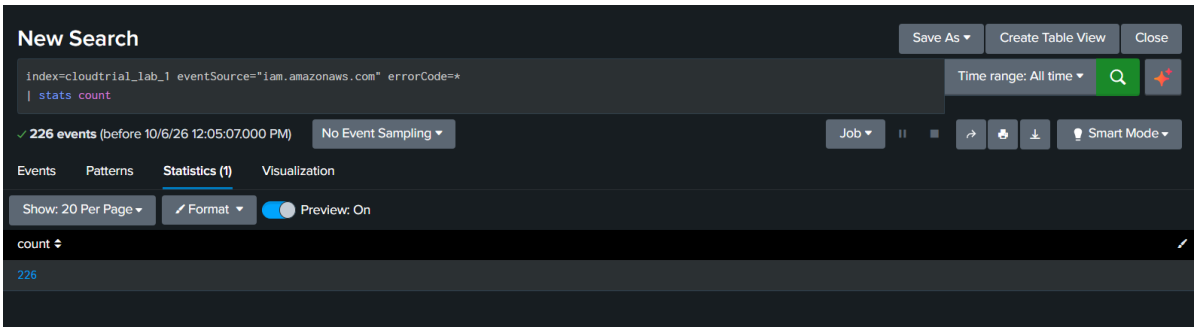
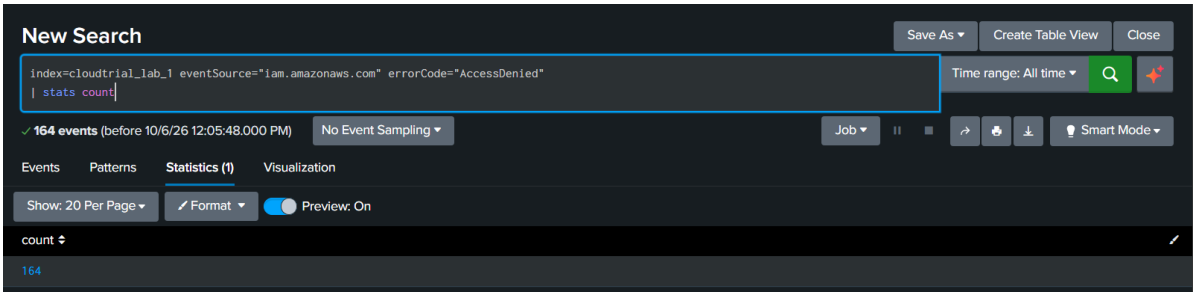


Figure C-8. Splunk AccessDenied event count: 164.



These Splunk screenshots provide visual evidence for the comparative counts reported in Section 6.7.

System/Component	**Configuration or Role**
Operating system	Windows 11 workstation
Elasticsearch	9.5.4
Kibana	9.5.4
AWS environment	Controlled course laboratory environment
AWS services examined	IAM and EC2
Logging service	CloudTrail
Analysis dataset	aws-Lab04 historical CloudTrail dataset
Automation framework	Python with Boto3